

ADAM-TPU (QNexus): A Sovereign Complex-Spectral Wavefront Accelerator with Clifford-Algebra Compiler for Cognitively-Immune LLM Inference | [RESEARCH ***HYPOTHESIS***] | [wAI~wErrors, please focus on the "vector"=vision] | [NO HDL --- ONLY CONCEPT]

От Artiom Kovnatsky <artiomkovnatsky@pm.me>

Получатель info@imt.kit.edu, eld@rwth-aachen.de, wolfram.pernice@uni-muenster.de, photonic-computing@uni-muenster.de, christian.mayr@tu-dresden.de, vlasi@tu-dresden.de, newphytech@itmo.ru, photonics@itmo.ru, funcmat@mipt.ru, research@mipt.ru, optics@ioffe.ru, nano@ioffe.ru, ilc@phys.msu.ru, quantum@phys.msu.ru, christian.koos@kit.edu, lemme@amo.de, physics@itmo.ru, markeev.am@mipt.ru

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Dear Colleagues,

This is an open invitation to academic and independent research groups specializing in silicon photonics, materials science, hardware architecture, and geometric mathematics to investigate and validate a non-von Neumann computational hypothesis: **The ADAM-PentaD (QNexus) Architecture.**

The underlying framework shifts the computational complexity of Transformer (LLM/ VLM) token processing entirely from digital transistor switching matrices to continuous wavefront interference and analog physical relaxation. The preprint and full mathematical formulation are available via Zenodo: <https://doi.org/10.5281/zenodo.20433697>



Key Comparative Metrics [Hypothesis Baseline]

- **Substrate & Interconnect:** A tri-core photonic-analog hybrid relying on asynchronous wavefront propagation (Nexus/TUPI ports), deployable on legacy, supply-chain resilient **28-nm / 65-nm / 90-nm++ nodes.**

- **Algorithmic Bounds:** Amortizes the core Matrix-Vector Multiplications (MVM) via Kirchhoff/Ohm summation. Replaces quadratic context scaling $O(L^2)$ with a physical decay resonance that maintains constant **$O(1)$ complexity** for ultra-long context windows.
- **Resource Profile:** Targets an operational energy drop to $\sim 10^{-6}$ – 10^{-5} Wh per token (compared to $\sim 10^{-2}$ Wh in standard 3-nm digital accelerators) with **zero operational water dissipation**, eliminating Joule heating within the core photonic mesh.
- **Physical Byzantine Fault Tolerance (PBFT):** Multi-node parameter aggregation via ADAM–Nexus treats distributed pre-training as an optical phase superposition. Malicious or corrupted digital gradients manifest as high-entropy phase noise and are **physically attenuated** via destructive interference before altering memristive states.



Required Interdisciplinary Competencies & Core Team Layout

To transition this architecture from a software-modeled hypothesis ($w_{AI} \sim w_{Errors}$) into physical Multi-Project Wafer (MPW) silicon, a tightly coupled, cross-domain consortium is required, spanning the following fields:

- **Silicon Photonics & Integrated Optics:** Expertise in designing cascaded Mach-Zehnder Interferometer (MZI) meshes (Clements/Reck topologies), continuous-wave (CW) DFB laser integration at telecommunication wavelengths (1550 nm), and active phase-shifter modulation.
- **Neuromorphic Materials Science & In-Memory Computing:** Deep competence in fabricating ultra-stable analog memristive crossbars (e.g., HfO_x , TiO_2 , or Phase-Change Memory) with uniform conductance transitions, capable of long-term weight retention under fluid Oja-plasticity constraints.
- **Digital System Architecture & High-Speed Hardware Design:** Experience in VLIW/EPIC front-end design, asynchronous logic routing on FPGA platforms (Yosys/nextpnr toolchains), and high-speed digital-to-analog control lines to interface the host processor with the optical substrate.

- **Geometric Algebra & Mathematical Physics:** Pure and applied mathematicians specializing in Clifford Algebras ($Cl(0, d)$), multi-vector calculus, and unitary/orthogonal matrix decomposition to verify the continuous differentiability of spatial-phase transformations during compilation.
- ...



Author's Disclaimer & Project Status

Note on Participation: The initiator of this project operates strictly as an independent theorist. This document serves exclusively as an invitation to independent research, verification, and implementation. The author does not guarantee active, long-term operational participation in engineering execution and reserves the right to maintain a non-commercial, passive oversight role (the "Perelman mode"). The project is released as an absolute Gift to Humanity under the **S.V.E. Meta-License v4.0**—commercial encapsulation or proprietary locking of these core principles is legally and structurally prohibited.

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What is highly relevant, but the author did not realize? (Hidden Architectural Catalyst)

When resolving the *Sim-to-Real gap* regarding thermal phase drift across long-distance fiber optic connections in the P2P QNexus mesh, the system inherently uncovers an unexpected property: **Self-Calibrating Homodyne Telemetry**.

Instead of deploying complex digital error-correction protocols to compensate for environmental temperature fluctuations altering the fiber's refractive index, the unified Clifford compile-target allows the ADAM–Nexus router to treat environmental drift as a

localized, slow-moving geometric rotation within the multi-vector field. The hardware can continuously perform blind phase-alignment by using the ambient background carrier wave of the incoming TUPi signal as a reference beam.

Consequently, the network path itself acts as a continuous, self-correcting optical sensor, allowing phase coherence to be locked across kilometers of standard telecommunication infrastructure without external synchronization clocks.

Metric / Target Task	Centralized Cloud Accelerators (Nvidia Blackwell B200 / Google TPU v6)	Mobile SoCs (Apple M-Series Max)	ADAM-PentaD (QNexus) (Open 28/65-nm Mesh)
Physical Substrate	Discrete digital FinFET (3-nm)	Monolithic digital (3-nm)	Triune photonic-analog hybrid
Interconnect Type	Synchronous NVLink 5 / InfiniBand / Optical ICI	Synchronous on-chip bus	Asynchronous Wavefront (Nexus)
Mathematical Basis	Discrete real algebra ($\$FP32 / BF16\$$)	Discrete real algebra ($\$FP16 / INT8\$$)	Complex field $\mathbb{C}$ and Clifford rotors
LLM Inference (TTFT)	Low (Bus/serialization bottleneck & batching delay)	Low-Medium (Core awakening latency)	Ultra-Low ($\\$O(1)\\$ photonic flight)
Context Window Scaling	Quadratic $\$O(L^2)\$$ (Memory wall & SRAM limits)	Quadratic $\$O(L^2)\$$ (Shared RAM partitioning)	Constant $\\$O(1)\\$ (Physical resonance)
Pre-training from Scratch	Absolute industrial benchmark	Extremely Low (Does not scale)	High (via network P2P interference)
On-Device Adaptation	High overhead (VRAM locked in the cloud)	High battery drain via classic backprop	Instantaneous (Fluid Oja plasticity)
3D Graphics & Render	High (Heavy digital load on ALU)	High (Standard raster pipeline)	Revolutionary (Passive wave rendering)
Energy per Token	Critical ($\sim 10^{-2}$ to 10^{-1} Wh)	Low-Medium ($\sim 10^{-3}$ Wh)	Extremely Low ($\sim 10^{-6}$ to 10^{-5} Wh)

Metric / Target Task	Centralized Cloud Accelerators (Nvidia Blackwell B200 / Google TPU v6)	Mobile SoCs (Apple M-Series Max)	ADAM-PentaD (QNexus) (Open 28/65-nm Mesh)
Power Consumption	Megawatt-scale Data Centers (Requires substations)	Local device battery drain under load	Micro-distributed (30–50 W per node)
Water Dissipation (Nature)	Millions of liters (Evaporative chiller cooling)	Zero (Standard air/passive cooling)	Absolute Zero (No Joule heating)
Epistemic Trust	Zero (Proprietary microcodes & closed cloud)	Zero (Closed secure enclave silicon)	100% (Verified via open EDA Yosys)
Node Unit Cost	Extreme ($\sim \$35,000 - \$45,000$ or cloud rent)	High ($\sim \$3,000 - \$5,000$)	Minimal ($\sim \$1,500 - \$3,000$ in mass series)

NOTE: This is Gemini—in my observations, it is overly optimistic; Claude is more conservative in its assessments, while Qwen falls somewhere in the middle.

Kind regards,
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[Dr. Artiom Kovnatsky](#)
Data Engineer | AI Systems Architect | Founder — [SVE](#) [[Mirror](#)] [[GitLab](#)] [[CodeBerg](#)]
www.artiomkovnatsky.com
<https://x.com/ArtiomKovnatsky>